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Joanna Gutierrez

EDUCATION	University of Texas Rio Grande Valley (UTRGV) <i>B.S., Computer Science, Minor: Electrical Engineering</i> Major GPA: 3.92/4.0 Secretary, Institute of Electrical and Electronics Engineers	Expected May 2027
RESEARCH	DUB REU , University of Washington	June. 2026 – Aug. 2026
EXPERIENCE	<i>Make4All Lab</i> Advisors: Kate Glazko, Jennifer Mankoff - Constructed a LangGraph-based agentic workflow for STL & SVG file generation using Python, ensuring fabricability of artifacts - Scraped up to 2,000+ fabrication-related posts across 16 Reddit communities & Bambu Lab Community Forum with BeautifulSoup & Playwright - Qualitatively analyzed 366 AI-use posts for fabrication across Bambu Lab forum & Reddit communities, identifying emerging themes: AI-assisted design, troubleshooting, & quality/criticism of AI outputs Undergraduate Researcher , UTRGV <i>Multiple Autonomous Robotic Systems Lab</i> Advisor: Qi Lu - Developed a custom genetic algorithm (GA) with Python for the Central Place Foraging Algorithm (CPFA) - Trained the GA to find the most optimal 7 CPFA parameters (resources collected x foraging time), with Gazebo & ROS2, 600+ training hours - Set up a Gazebo, ROS2/Ardupilot SITL development environment for dual-drone collaboration enabled with April-Tag detection Robots In The Real World REU , Oregon State University <i>Social Haptics & Embodiment Lab</i> Advisors: Antara Shah, Naomi Fitter - Extended & prototyped a multi-modal (touch & speech) trivia interface for older adults with Python/QThread/PyQt; designed robot behaviors & audio cues (JSON, AWS Polly) - Co-led 8 user-studies (2x2 reward x difficulty design) on a robot-hosted trivia activity for older-adult cognitive & social wellbeing - Conducted post-study interviews & analyzed RoSAS, GAToRS, accuracy, & response-time data across 4 conditions; authored a technical writeup on our findings [PDF]	Oct. 2025 – Aug. 2026 June. 2025 – Aug. 2025
PUBLICATIONS	<i>P. Li, J. Gutierrez, F. Hernandez, Q. Lu, L. Zhou.</i> “LLM-Foraging: Large Language Models for Decentralized Swarm Robot	

Foraging.”

To appear in Proceedings of Distributed Autonomous Robotic Systems (DARS 2026), Oct. 2026, Tokyo, Japan. [Accepted] Preprint: <https://arxiv.org/abs/2605.01461>

J. Becerril, M. Vargas, J. Herrera, J. Gutierrez, J. Rios, M. Amjadian, C. Tarawneh, J. Yang, Q. Lu.

“Non-Contact Vibration-Based Damage Detection of Civil Structures Using a Cost-Effective Autonomous UAV.”

In Proceedings of International Conference on Unmanned Aircraft Systems (ICUAS 2026), June 2026, Corfu, Greece. <https://doi.org/10.1109/ICUAS69441.2026.11598690>

TEACHING	Teaching Assistant	
EXPERIENCE	Data Structures & Algorithms – Qi Lu (3333-01)	Fall 2026
	Data Mining – Li Zhang (6366-01)	Spring 2026
	Discrete Mathematics – Li Zhang (3310-01, 3310-04, 3310-05)	Fall 2025 – Spring 2026
	Peer Led Team Leader	
	Calculus 1 – Joseph O. Alanis (2413-07P)	Fall 2024
	Calculus 1 – Rodrigo H. Pena Jr. (2413-09P)	Spring 2024
FELLOWSHIPS & AWARDS	Break Through Tech (BTT) AI/ML Fellow	May. 2026 – Present
	- Selected as 1/1k fellows nationwide out of 4k+ applicants for a highly competitive year long AI/ML program - Awarded fully-funded enrollment in eCornell’s Machine Learning Foundations certificate program by Cornell University	
	Keppel AmFels Scholarship Recipient	Fall 2026
	Robert L. Townsend Scholarship Recipient	Fall 2026
	Stars Scholarship Recipient	Fall 2025
	Last Mile REU Fund Recipient	Summer 2025
SELECTED PROJECTS	Natural Language Task Specification for Autonomous UAV Detection & Tracking	Jan. 2026 - May 2026
	<i>Senior Capstone Project</i>	
	- Built a text-prompt-driven UAV object detection system, integrating ChatGPT & YOLOv8m-world with ROS2/ArduPilot SITL in Gazebo via a custom PyQt interface & UDP protocol - Co-authored a technical paper showing LLM & CV bundling can outperform traditional CV constraints, enabling a UAV to interpret & act on abstract natural language without additional training [PDF]	
	Burnout Bots	November 2025
	<i>HackR Hackathon 2025</i>	
	Designed & validated logic-gate gadgets (NAND & fan-out) using burnt-out robots as	

pathway obstacles in a tilt-based grid model, proving universal computation under a novel limited movement constraint

- Co-authored a technical research paper detailing findings; selected as winner & included in the official event proceedings compiled by Dr. Tim Wylie [[PDF](#) / [2025 Proceedings](#)]

TECHNICAL
SKILLS

Languages Python, C++, C, Dart, R, SQL

Technologies Git, Docker, Jupyter Notebook, Google Colab, Visual Studio

Developer Tools ROS2, Vite, ReactJS, Flutter, RStudio, MySQL